Javad Komijani

Computational Scientist & ML Researcher

Zurich, Switzerland
+41 76 796 9992
jkomijani@ethz.ch
jkomijani.github.io
jkomijani

Professional Summary

Computational scientist and machine learning researcher with 10+ years of experience in scientific computing and generative modeling. Specialized in PyTorch-based diffusion models, normalizing flows, and differentiable programming. Strong background in applied mathematics, statistical modeling, and scientific software engineering.

Technical Skills

Programming	Python, C++, Cython, Bash
ML/AI	PyTorch, Diffusion Models, Normalizing Flows, Generative Modeling, Distributed Training
Mathematics	Differential Equations, Linear Algebra, Optimization, Statistical Modeling
Tools	Linux, Git, SQL, Pandas
Scientific	NumPy, SciPy, Scikit-learn, HPC, Parallel Computing, Monte Carlo Methods

Professional Experience

2021–Present **Senior Postdoctoral Researcher**, *ETH Zurich*

- Introduced a novel score-matching framework for diffusion models on Lie groups.
- Designed and implemented GPU-based generative models for large-scale lattice simulations.
- Developed installable PyTorch extension packages for differentiable linear algebra and ODE solvers.
- Implemented distributed GPU training and large-scale numerical simulation workflows on HPC clusters.
- Developed and contributed ML modules to the interTwin (EU interdisciplinary Digital Twin Engine).
- Supervised PhD and Master's students and taught quantum mechanics.

2015–2021 **Postdoctoral Researcher**, *TU Munich / Glasgow / Tehran*

- Developed models and applied Bayesian inference & Monte Carlo simulations to large-scale datasets.
- Developed scalable lattice-QCD simulation pipelines in HPC environments.
- Collaborated internationally in computational and mathematical physics projects.

Open-Source Scientific ML Packages

normflow	Installable PyTorch package for normalizing flows on Lie groups & lattice gauge theory.
diffusion	Installable PyTorch package for diffusion-based generative modeling on Lie groups & gauge theories.
torch_linalg	Installable PyTorch package for differentiable SVD and eigendecomposition of unitary matrices.
torch_solve	Installable PyTorch package implementing adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*

2009 **MSc in Electrical Engineering**, *University of Tehran*

Publications

A complete list of over 25 peer-reviewed publications can be found at this link.

Languages

English	Fluent
German	Elementary